Week 50 Staircases and lengths

Facilitator guide | Final student packets v2 | Prepared 3 October 2026 | Revised 23 October 2026 | Unpiloted

Mathematical overview

Length invariant. Every finite horizontal/vertical path from S=(0,0) to F=(160,120), using only right and up moves, has horizontal total 160 mm and vertical total 120 mm. Its length is therefore **280 mm**, regardless of the number or sizes of its pieces. Zero-length pieces do not create turns; adjacent collinear pieces should be merged when counting.

The straight diagonal has length **200 mm**, since $160^2+120^2=200^2$. Staircases can be made arbitrarily close to it while retaining length 280 mm. Here “close” means the largest **vertical** gap $|y - 3x/4|$ over the whole path, not the length along the route or a perpendicular distance. Closeness of curves does not imply closeness of lengths. No finite staircase becomes the diagonal.

If leftward travel is allowed but downward travel is not, let L be the total distance traveled left. Rightward travel must be 160+L, so total length is **280+2L mm**. This counts repeated traversal as traveled length; it is not the area or length of the inked set.

Fewest turns. In Problem 4's strip (15 mm above and below the diagonal, boundary included) the fewest turns is **8**, and page 2 has a witness. A horizontal piece spans at most 40 mm, or 20 mm if it starts at S or ends at F, so seven turns cover at most $20+3\times40=140$ of the 160 mm. Without the boundary the answer is 9. In the 7.5 mm strip of Grades 4-5 Problem 5 it is **16**.

Discovery by band. K-1 compares string lengths and makes close or backtracking routes. Grades 2-3 explains totals and minimum turns. Grades 4-5 separates approximation of location from approximation of length. Measurement suggests equality; regrouping all horizontal and vertical pieces proves it.

Prepare and launch

Per pair: student pages, pencils, a ruler, two nonstretch strings at least 40 cm long, and removable tape. Print at actual size. Main rectangles are **160 mm wide by 120 mm high**; verify the printed 100 mm calibration before using string or a ruler. The diagram on the next guide page is deliberately smaller, not a measuring mat. For K-1, a 30 cm pipe cleaner can replace string: it holds its corners and leaves about a 2 cm tail past F on every staircase (untried).

Readiness. K-1 needs right/up directions and side-by-side comparisons, with adult reading. Older entry requires addition of lengths; explaining the eight-turn lower bound needs control of a vertical-gap constraint. Children need not derive a square-root formula. Keep the three adult-led tables and offer the minimum-turn problem as exploration before proof.

Launch, 4 minutes. Let children bend string into a path, then follow the printed 40-by-30 mm example piece by piece, A to D, and straighten it beside the 70 mm bar. Show that string must follow every corner rather than cut across it. Ask, “Can you make a very different route between the same two dots?”

First-visit hour. 0-5 running around; 5-9 launch; 9-48 Problems 1-3, then Problem 4 for any pair that finishes; 48-55 share; 55-60 reset. Save a route before changing its rules.

Key for the common path problems

Problem 1, all bands. Any two different right/up routes between the printed endpoints work. For example, right 160 then up 120, and up 60, right 80, up 60, right 80. Each is 280 mm. Small differences in measured strings are measurement error, not a counterexample to the segment-total proof.

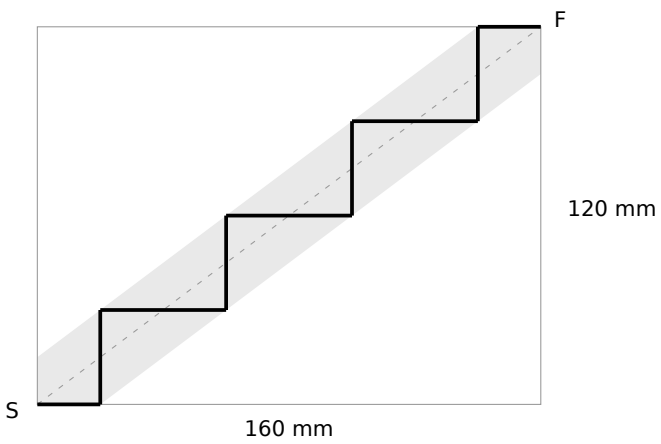
Problem 2, all bands. Both solid and dashed routes are **280 mm**. Solid: 160+120. Dashed: 60+80+60+80. K-1 should circle “same length.” The two paths need not look equally direct for their horizontal and vertical totals to agree.

Problem 3, all bands. The given four-step staircase reaches a greatest vertical gap of **30 mm**: its lower corners lie outside the strip, so any staircase inside the strip is closer to the diagonal. One improvement is eight repetitions of right 20, up 15, whose greatest gap is **15 mm**; it stays within the gray strip, including its boundary. Both staircases have length **280 mm**, whereas the diagonal is **200 mm**. Along each segment the signed gap is linear, so checking its endpoints controls the whole segment.

The eight turn optimum

Problem 4, all bands. The fewest turns is **8**. A boundary-following witness, in millimeters from S, is
(0,0), (20,0), (20,30), (60,30), (60,60),
(100,60), (100,90), (140,90), (140,120), (160,120).
Its nine nonzero segments make eight direction changes. All vertices have vertical gap in [-15,15]; linearity gives the same for the full path.

Guide diagram
not to scale



Why seven turns cannot work. Inside the strip, a horizontal segment can span at most 40 mm: crossing a vertical-gap range of 30 takes 30/(3/4)=40. A horizontal segment beginning at S or ending at F can span at most 20 mm. With at most seven turns, there are at most eight alternating horizontal/vertical segments. If there are four horizontal segments, at least one is an endpoint segment, so horizontal travel is at most 20+3×40=140 mm; if there are fewer, it is at most 120 mm. Neither reaches the required 160 mm.

Key for changed rules and narrower strips

K-1 Problem 5; Grades 2-3 Problem 5. Here are two distinct 360 mm routes, given as right (R), left (L), and up (U) distances in millimeters:
Route A: R80, U40, L40, U40, R120, U40.
Route B: R120, U90, L40, U30, R80.
Both stay in the rectangle, end at F and travel left a total of 40 mm. Each length is 280+2×40=**360 mm**. K-1 needs only two equal longer paths, so these give a complete witness without requiring arithmetic.

K-1 Problem 6; remainder of Grades 2-3 Problem 5. A positive leftward piece cannot make a path shorter than 280 mm. Every millimeter left must be recovered by an extra millimeter right, adding two millimeters to the baseline. A route with any positive leftward travel is strictly longer than a staircase.

Grades 4-5 Problem 5. The strip is 7.5 mm, half of Problem 4's. The fewest turns is **16**: start U7.5, then use eight R20 pieces with U15 between consecutive ones, and finish U7.5. Its greatest vertical gap is exactly 7.5 mm, so it stays in the strip, boundary included. **Why 15 turns cannot work:** here a horizontal piece spans at most 20 mm, and at most 10 mm if it starts at S or ends at F. With 15 or fewer turns there are at most 16 pieces, so at most 8 horizontal pieces; if there are 8, one is an end piece, so they cover at most 10+7×20=150 mm, and fewer cover at most 140 mm, short of 160. The staircase is still **280 mm**, the same as any staircase in the wider strip, so it is not shorter.

Grades 4-5 Problem 6. The printed left-once route is Route A above. Horizontal travel is 80+40+120=240 mm and vertical travel is 40+40+40=120 mm, giving **360 mm**. Net horizontal displacement remains 160, but horizontal distance traveled is no longer 160; this is the exact part of the staircase argument that changes.

Grades 4-5 Problem 7. No finite right/up staircase can have length 200 mm. Its total horizontal and vertical distances remain 160 and 120. With n equal right-then-up steps, the largest vertical gap is 120/n mm, which can be made smaller than any specified positive tolerance by taking a sufficiently large finite n. The length remains 280 mm. The diagonal is a separate limiting curve with length 200 mm, not a last member of the staircase sequence.

Optional hints and facilitation checks

For length, suggest collecting the right pieces and the up pieces separately; the opening example keeps them in path order. For a strip violation, compare vertically with the diagonal at the same x-position. For minimum turns, ask how far one horizontal piece could possibly go before leaving the strip; do not begin by supplying the optimal coordinates.

Let younger children improve a legal route and count its turns without requiring the lower-bound proof. If string measurements seem unequal, check for stretch, shortcutting corners or inconsistent endpoints. The exact argument is about ideal centerline lengths; printed stroke thickness is not an extra corridor allowance.

Source and verification

John D. Norton, *Badly Behaved Curves*, sections “Proof that √2=2,” “The Limit Curve” and “Badly Behaved Limits of Curves”: author’s exposition. The 160-by-120 mm staircase version, exact finite corridor witnesses, eight-turn lower bound and backtracking examples here were independently derived and checked. All printed dimensions were checked digitally; string handling and session timing are unpiloted proposals, not reports of physical tests.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

First visits in all bands: P1-3, then P4 for any pair that finishes, as in the hour on page 1. On return, K-1 can explore fewest turns (P4) or add leftward travel (P5-6). Grades 2-3 can justify the turn minimum (P4) or construct fixed longer paths (P5). Grades 4-5 can separate the minimum-turn proof (P4), the narrower strip and its turn count (P5), and leftward travel/finite approximation questions (P6-7). Save actual-size route records and keep location closeness distinct from traveled length.